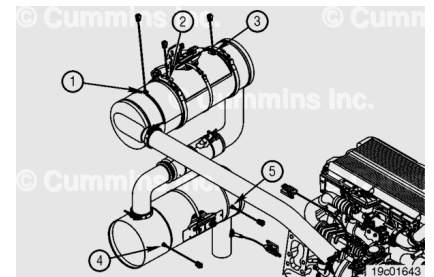


Trainings ▾

General Information

Exhaust gas temperature sensors are used to measure the temperature of exhaust gases for use with the aftertreatment system.

The aftertreatment temperature sensors are located in the aftertreatment system. The aftertreatment diesel oxidation catalyst intake temperature sensor (1) is located closest to the turbocharger outlet. The aftertreatment diesel particulate filter intake temperature sensor (2) is located between the aftertreatment diesel oxidation catalyst and the aftertreatment diesel particulate filter. The aftertreatment diesel particulate filter outlet temperature sensor (3) is located in the outlet of the aftertreatment diesel particulate filter. The aftertreatment selective catalytic reduction (SCR) catalyst intake temperature sensor (4) is located in the intake of the aftertreatment SCR catalyst. The aftertreatment SCR catalyst outlet temperature sensor (5) is located in the outlet of the aftertreatment SCR catalyst.



Note : The sensors have different connectors and lengths to prevent them from being installed incorrectly. Take note of the location of each sensor before removing it from the aftertreatment system.

Due to the number of various aftertreatment configurations, this procedure has been written to be generic. **Not** all illustrations within this procedure will represent the application that is being serviced.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,

remove the negative (-) battery cable first and attach the negative (-) battery cable last.

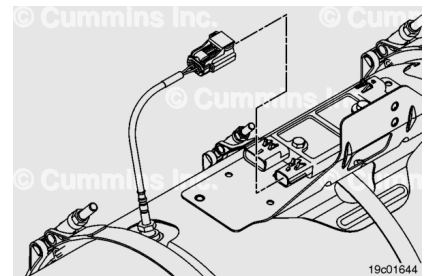
- Disconnect the batteries. Refer to the OEM service manual.



Remove

⚠ CAUTION ⚠

If the temperature sensor wire connectors are not connected to the proper locations after installation, aftertreatment system damage can result.



Note : Label each of the wire connectors prior to disconnecting the temperature sensors from the appropriate aftertreatment temperature sensor interface module.

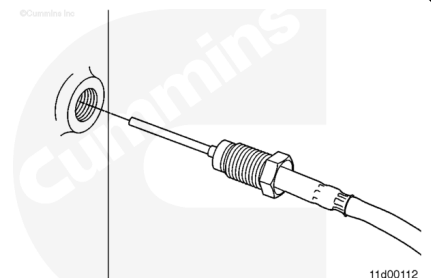
Slide the locking tab on the top side of the connector to the unlock position.

Press down on the tab near the end of the sensor connector, where the harness wires lead into the connector.

Pull outward on the connector, away from the interface module, to disconnect it.

⚠ WARNING ⚠

The aftertreatment system will stay hot to the touch for long periods of time after the engine has been shut down. To reduce the possibility of personal injury, avoid direct contact of hot components with your skin.



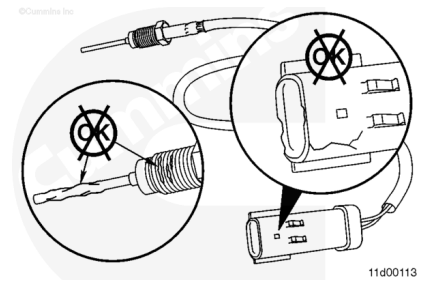
Loosen the retaining nut and remove the aftertreatment temperature sensor from the aftertreatment system.

Inspect for Reuse

Inspect the aftertreatment gas temperature sensors for damaged or exposed wires, bent or broken pins, damaged connectors, or damaged threads.

Inspect the tip of the aftertreatment gas temperature sensor for cracks, dents, kinks, and carbon buildup.

Replace the sensor if damage is found.



Install

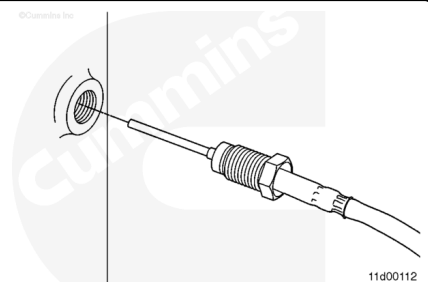
Apply a coating of Loctite™ 80209, 51002, 76732 Copper or Silver Grade anti-seize, or equivalent, to the exhaust gas temperature sensor threads prior to assembly.

Install the aftertreatment gas temperature sensor.

Note : Be sure to install the aftertreatment gas temperature sensor in the same location from which it was removed.

Tighten the nut that secures the sensors to the aftertreatment system.

Torque Value: 30 N·m [22 ft-lb]



⚠ CAUTION ⚠

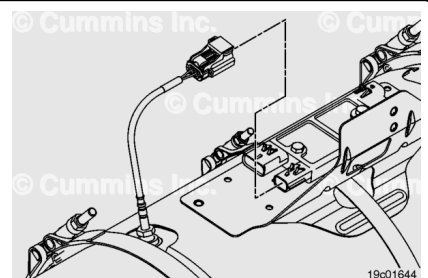
If the temperature sensor wire connectors are not connected to the proper locations after installation, aftertreatment system damage can result.

Make sure the aftertreatment gas temperature sensors are connected to the correct positions on the aftertreatment temperature sensor interface module.

Connect the aftertreatment gas temperature sensor connections, as noted during removal.

Push the connectors together until they lock. An audible click will be heard when the connector locks in place.

Slide the locking tab to the locked position.



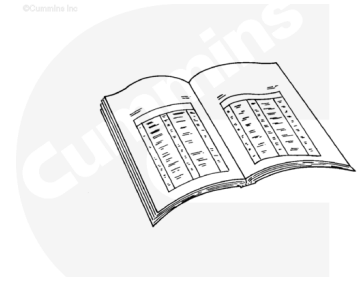
Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for proper operation.

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